

Decoding Capital Flow in Climate Finance: A Machine Learning Analysis of Global Climate Finance Per Capita

Using a panel of 117 countries (2000-2023), this study applies supervised machine learning to test if Need and Readiness factors predicts the allocation of International finance. Random forest and XGBoost models reveal need factors explains only 18% of the capital flow variance.

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01 - Introduction

Why does Climate finance allocation matter?
Despite developed country pledges, climate finance allocation remain disproportionate, with middle-income countries receiving greater funding than more vulnerable regions such as Sub-Saharan Africa and South Asia.

Global climate finance	Annual Losses	Global Exposure
\$1.2T	\$2.4B	80%
2023 total	EM-DAT 2026	High & medium risk

02 – Research question

Does objective climate vulnerability predict finance allocation?
Equitable climate finance should reflect adaptation and mitigation needs. Yet trends in finance flows indicate a weak alignment with objective measures of vulnerability.

Climate vulnerability shows a weak relationship with finance allocation, indicating a misalignment in global climate finance.

03 – Data sources

- Panel dataset: 117 countries, 2000-2023
- ND-GAIN country index – vulnerability and readiness
 - World Bank Open Data – GDP, population, CO₂ per capita
 - EM-DAT (CRED) – disaster economic losses by country-year
 - OECD (CRDF) – Climate related development finance by country-year

All sources are open access. Merged on ISO 3166 country-codes.
ND-GAIN scores averaged across all available years for robustness

04 - Methodology

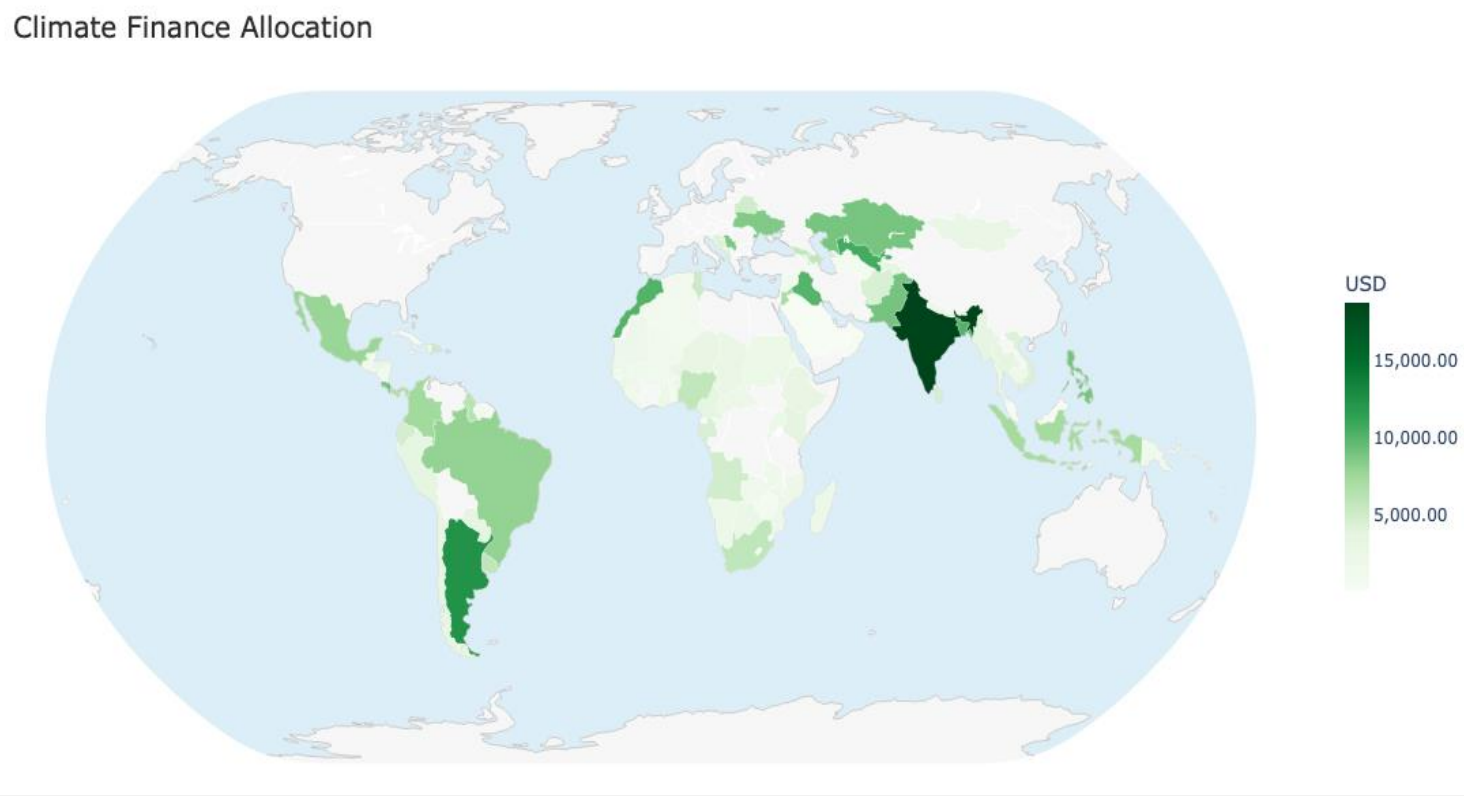
- Supervised Machine Learning
- Target: log(climate finance per capita)
 - 7 features: physical risk, population, CO₂, GDP_PC, damages_PC, Inflation(CPI), and institutional quality
 - Models: OLS • Random Forest • XGBoost
 - Evaluation: R²
 - Importance: Shap Explainer Waterfall plot



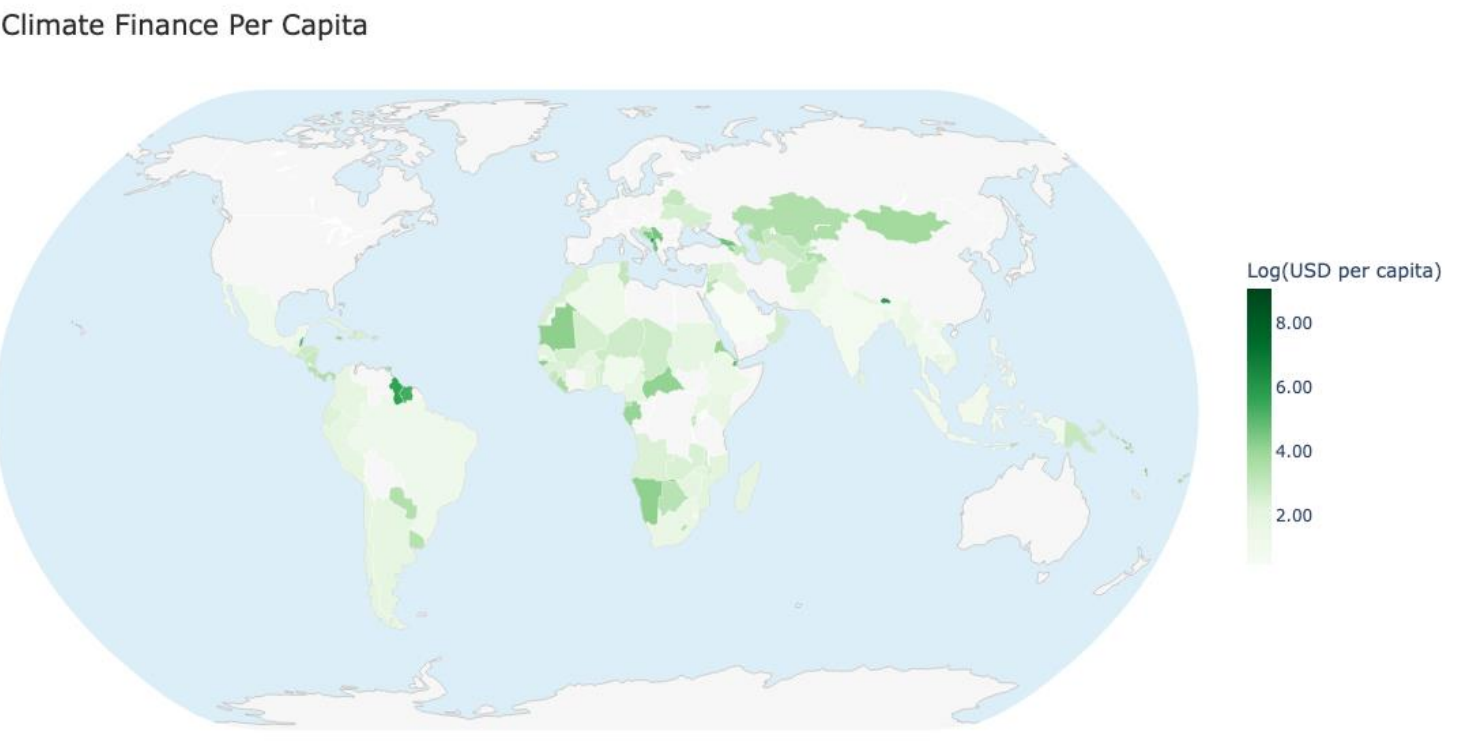
Key finding: Machine Learning Shows Vulnerability and Risk Plays Limited Role in Finance Allocation
XGBoost outperforms OLR and Random Forest; population, emissions profile, and income are key predictors • N = 117 country-year observations • 2000-2023

05 – Where the money goes

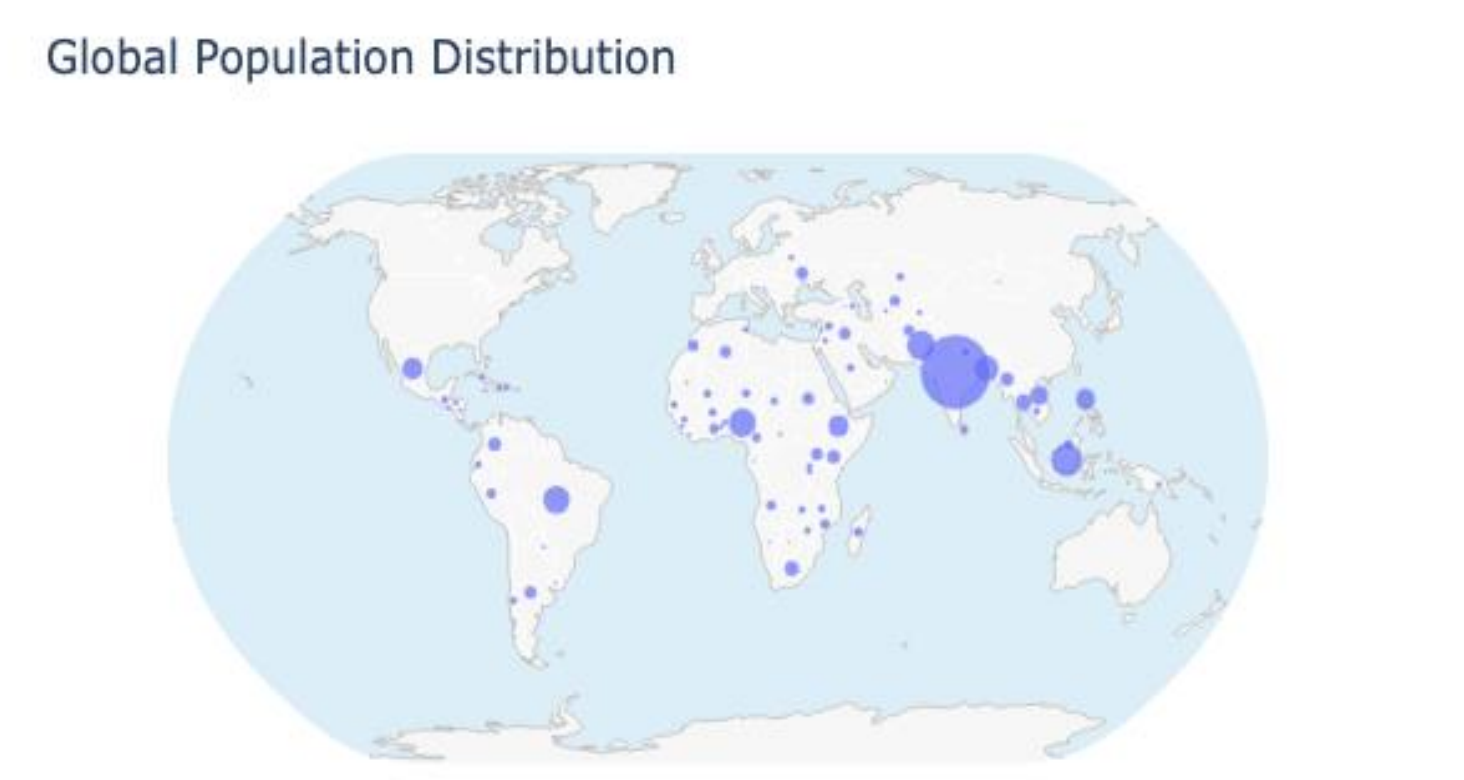
Geographical concentration of climate finance



Climate finance is concentrated in South Asia, Europe, and South America.



Population adjustment reveals higher relative allocations to Africa.



Perceived equity depends strongly on the metric used.

06 – Where the risk is

Highest risk concentrated in the global south



Sub-Saharan Africa, South Asia, and small island developing states carry the highest composite hazard scores for flood, heat, and drought.

09 – Model Performance

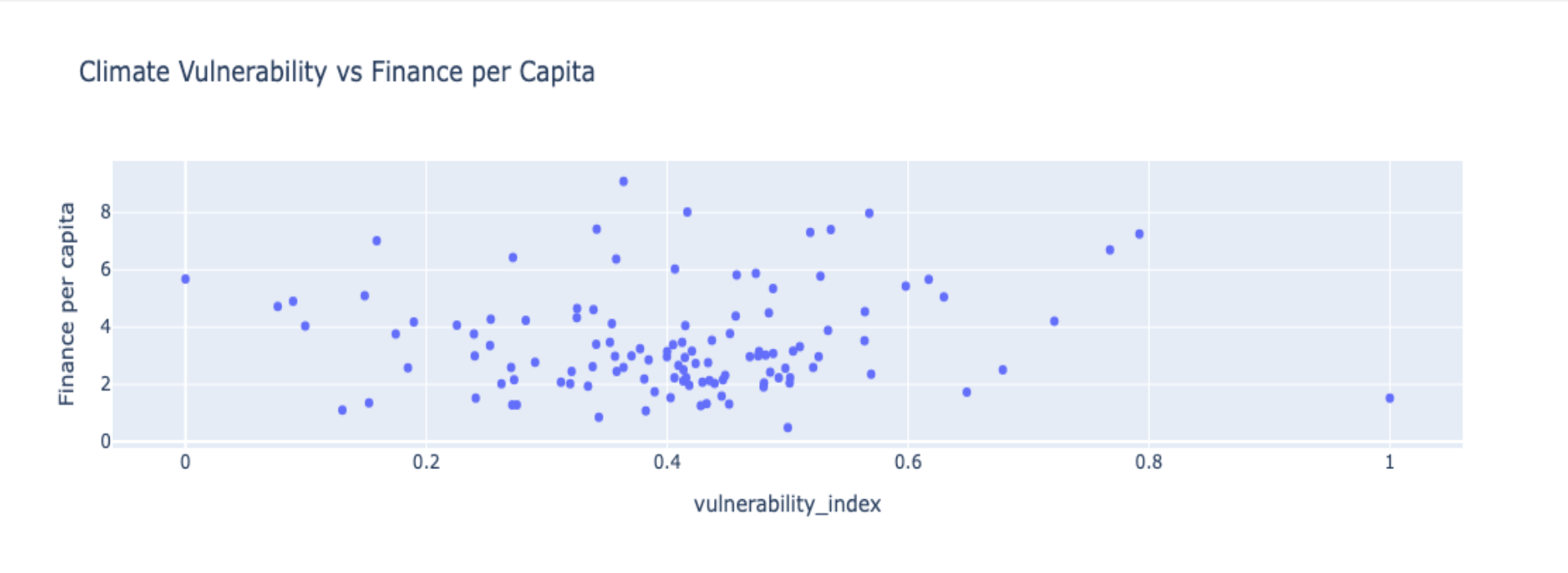
XGBoost achieves highest explanatory power

Model performance comparison — R ² score (%)	
Model	R2 Score
Linear Regression	14.26%
Random Forest	17.52%
XGBoost	18.39%

Minimal performance gains from **non-linear models** imply that unobserved **political and institutional dynamics**, rather than **physical vulnerability**, primarily shape allocation outcomes.

07 – The mismatch

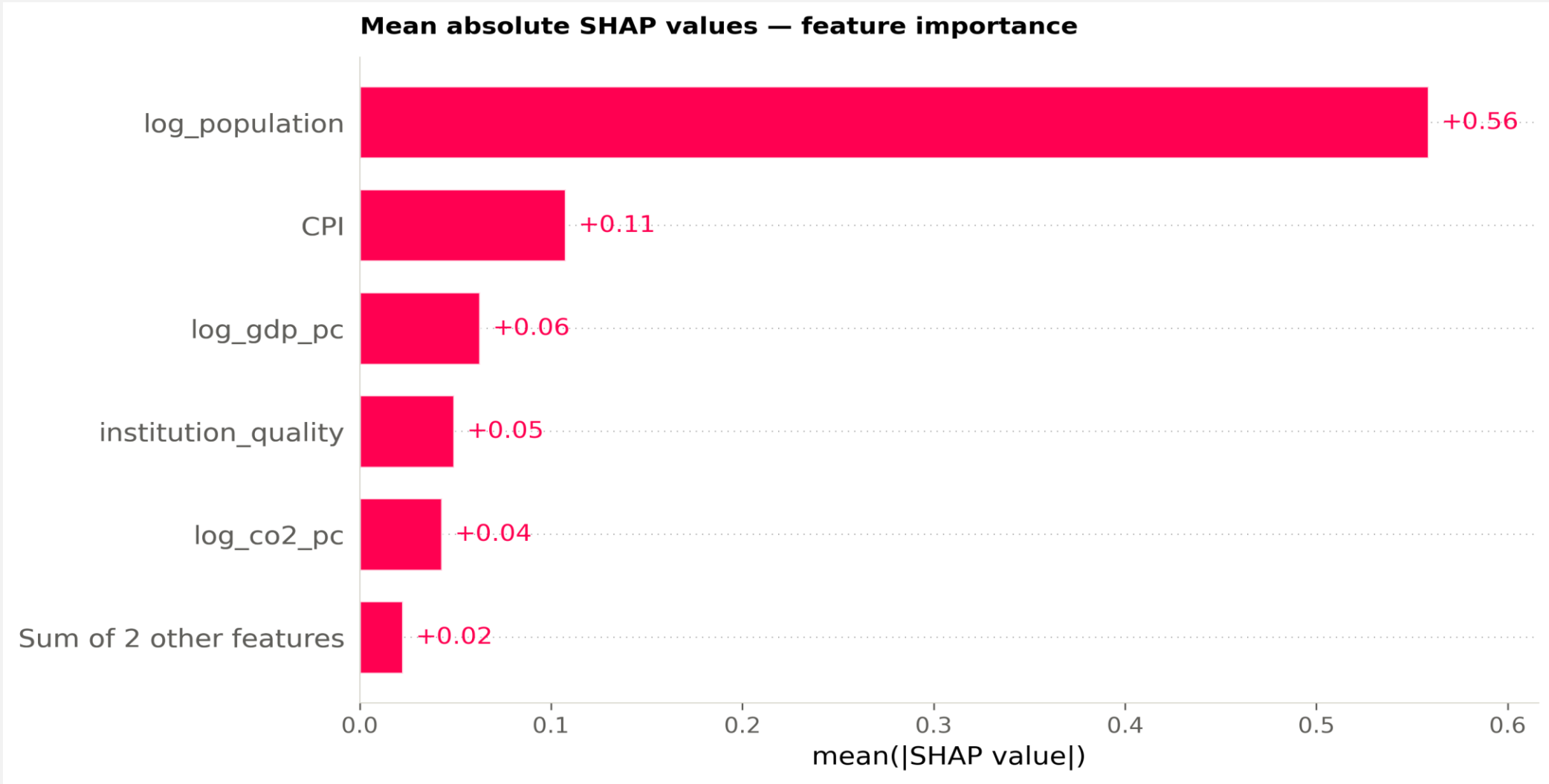
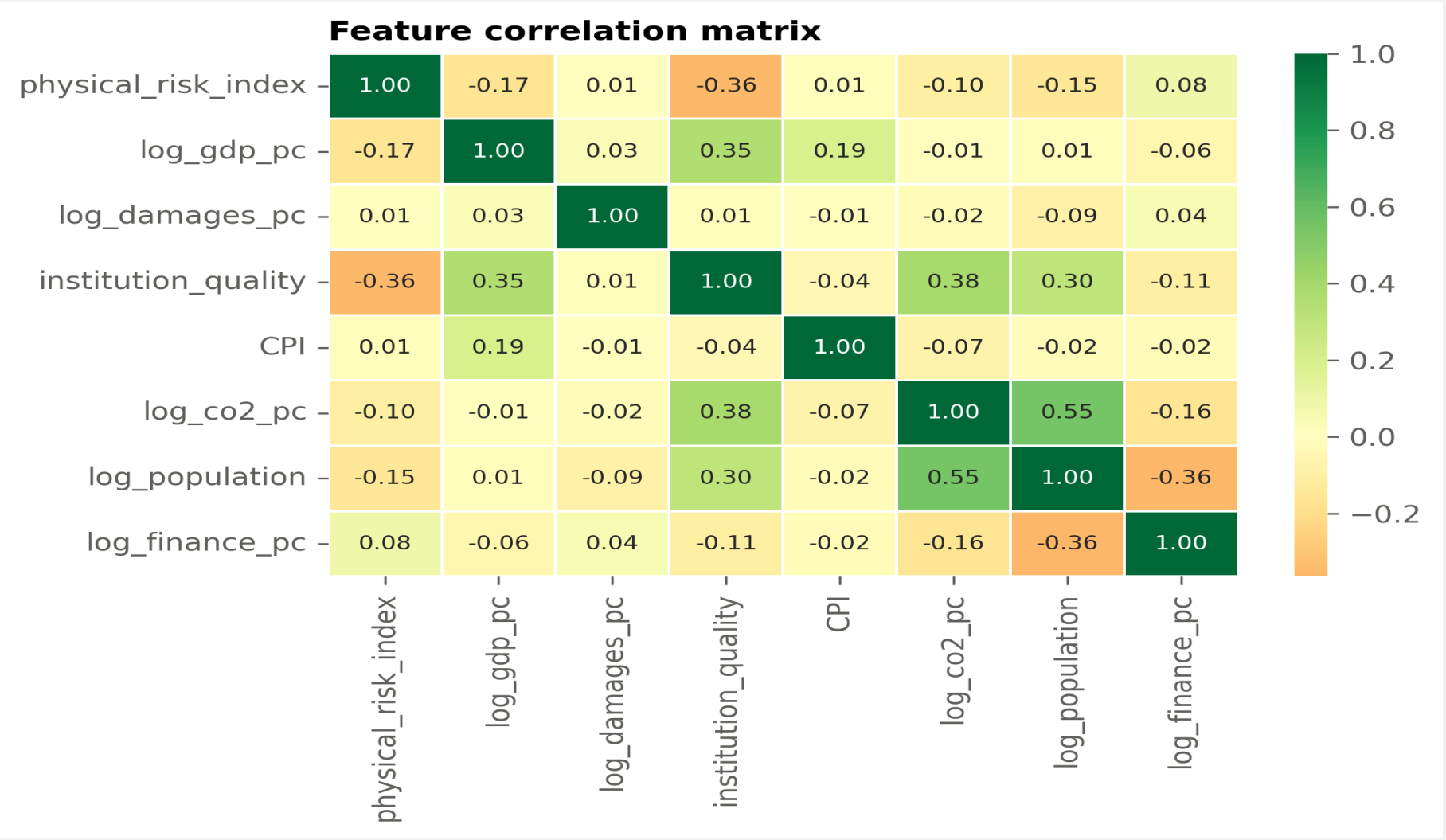
High-risk nations receive less finance



Several **low to medium risk** countries, such as **Antigua and Barbuda**, receive substantially higher mitigation and adaptation finance than **higher-risk countries**, including **Bangladesh and Mauritania**.

10 – Feature Importance – core ML result

Within the limited share of climate finance allocation explained by the **model (18%)**, **population** is the dominant predictor, whereas **physical climate vulnerability** plays a **minimal role (2%)**, indicating a systematic misalignment between finance flows and objective climate risk.



Data & Methods

- Open-source data were sourced from the World Bank, OECD, EM-DAT, ND-GAIN, and the Climate Policy Initiative.
- Datasets were merged by country code, country name, and year.
- Missing GDP and population data were initially handled using linear interpolation; observations with insufficient data for interpolation were excluded.
- Total damages and adaptation and mitigation finance were imputed as zero where no records existed, acknowledging potential model bias. Climate risk was defined as the normalised average of vulnerability and (1 – readiness). Institutional quality was measured as the average of rule of law and control of corruption.
- GDP, finance allocations, and damages were converted to per-capita values and log-transformed to reduce skewness.
- Target variable: log climate finance per capita.

Limitations

- Climate Policy Initiative (CPI) data do not provide country-level finance breakdowns, as reporting is largely regional; analysis was therefore restricted to OECD (CRDF) finance data.
- Annual total damage data contain substantial missing values across countries.
- Finance allocations are missing or underreported for several higher-income countries, including the UK, US, Canada, Australia, and China.
- Climate finance aggregates heterogeneous instruments (grants, loans, equity) into a single measure, despite differing allocation logics that models cannot distinguish.
- This study is observational; estimated associations cannot be interpreted as causal.
- Unobserved factors—such as colonial history, donor–recipient trade relationships, or participation in international climate finance frameworks—may influence allocation patterns attributed to GDP or climate risk.

Future work

- Expand disaster damage coverage using alternative sources such as Swiss Re Institute natural catastrophe data to improve robustness of damage estimates.
- Apply machine-learning methods to further identify strategic and political drivers influencing climate finance allocation and wealth effects.
- Separately model public and private climate finance to test whether public finance responds more strongly to physical climate risk than return-driven private capital.
- Examine how different financing purposes (e.g. adaptation vs mitigation) reflect donor priorities and global climate finance governance frameworks.
- Assess how post-COVID-19 and conflict-related economic shocks have altered climate finance sources, volumes, and allocation patterns.

References

- Global Landscape of Climate Finance (GLCF), 2023
- ND-Gain Country Index, 2025
- Notre Dame University, 2023
- World Bank Open Data,. 2024
- EM-DAT – The international disaster database, 2024
- OECD – Climate and Development Finance Data, 2023